

COMSATS University Islamabad, Sahiwal Campus

SOFTWARE REQUIREMENTS SPECIFICATION

(SRS DOCUMENT)

for

NeuraPath

Version 1.0

By

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Revision History

Name	Date	Reason for changes	Version

Application Evaluation History

Comments (by committee) ***include the ones given at scope time both in doc and presentation**	**Action Taken**

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Signature _____

1 Introduction

NeuraPath is an **AI-powered career guidance platform** designed to assist students, job seekers, and professionals in making informed career decisions. The system leverages **Machine Learning (ML)**, **Natural Language Processing (NLP)**, and **real-time job market data** to provide personalized AI Career Intelligence Engines, Smart Resume Enhancer, and skill gap analysis.

This document outlines the system's **scope, objectives, features, architecture, and constraints**, ensuring a shared understanding among all parties involved. It is structured to help readers easily navigate specific requirements such as functional specifications, performance goals, usability considerations, and design constraints.

Furthermore, this SRS will act as a **baseline throughout the Software Development Life Cycle (SDLC)**, guiding activities including design, implementation, testing, and future maintenance. It also serves as a **formal agreement** between the client and the development team, ensuring that the delivered system meets user expectations and project objectives.

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1.1 Purpose

The purpose of this document is to define the requirements of **NeuraPath**, an AI- powered mobile application designed to provide intelligent career guidance for three major user groups: **students, job seekers, and professionals**.

The system addresses challenges such as lack of personalized guidance, high ATS resume rejection rates, and unclear career progression. Using **ML, NLP,** and real-time labor market data, it offers AI Career Intelligence Engines, Smart Resume Enhancer, job matching, and Growth Pathway Analyzer. An admin dashboard is also included for monitoring and analytics. This document ensures a shared understanding of system objectives and serves as the foundation for design and development.

1.2 Scope

1.2.1 In-Scope

- **AI Career Intelligence Engine**: ML-based AI Career Intelligence Engines and skill gap analysis.
- **Smart Resume Enhancer**: ATS compliance, keyword enhancement, and NLP-based resume parsing.
- **AI Job Matcher**: Real-time matching with scraped market data.

- **Growth Pathway Analyzer**: Promotion roadmaps, salary benchmarks, and growth pathways.
- **Skill Insight & Learning Advisor**: Targeted courses and certifications to improve employability.

1.2.2 Out-of-Scope

- Direct job application submission (system only recommends opportunities).
- Psychological or personality-based career assessments.
- Coverage of highly niche or emerging industries with limited market data.

2 Overall Description

2.1 Product perspective

NeuraPath is an **entirely new product** designed as a mobile application to provide intelligent career guidance. Unlike existing tools such as **LinkedIn Career Explorer, MyPlan**, or Resume **Worded,** which offer only fragmented solutions (e.g., career suggestions without Smart Resume Enhancer, or resume tools without career growth features), NeuraPath unifies all major career development functionalities into a single platform.

2.2 Operating environment

- The system shall operate as a **mobile application** on Android (version 10 and above).
- Developed using **Flutter (Dart 3.0)** with **GetX** for state management.
- Backend services hosted on **Firebase** (Authentication, Firestore, ML Kit) [2] and **Cloudinary.[1]**
- Web scraping modules run on **Beautiful Soup/Selenium** to fetch real-time job data.
- Internet connectivity (Wi-Fi/4G/5G) is required for real-time features.

2.3 Design and implementation constraints

CO-1: The system shall be developed using **Flutter (Dart 3.0)** as the primary cross-platform development framework to ensure compatibility across Android and future web versions.

CO-2: The backend and authentication services shall be implemented using **Firebase**, including **Firestore**, **Firebase Auth**, and **ML Kit** for real-time data handling and AI features.

CO-3: The system shall utilize **GetX (v4.6.5)** as the state management library for efficient UI updates and reactive data binding.

CO-4: The system shall use **Cloudinary API** for secure and scalable image storage and retrieval.

CO-5: The system shall integrate **BeautifulSoup/Selenium** (Python) for job market web scraping, limited to supported external platforms.

CO-6: The application shall require an **active internet connection** to access real-time job data, Smart Resume Enhancer, and recommendation services.

CO-7: The system's data accuracy and recommendations are constrained by the **availability and reliability of external APIs** and job data sources (LinkedIn, Rozee.pk, etc.).

3 Requirement identifying technique

This section describes the requirements identifying technique(s) which further help to derive functional requirements specification. The selection of the technique(s) will depend on the type of project.

3.1 Use case diagram

The Use Case Diagram illustrates the main interactions between users and the NeuraPath system. Three primary actors—Student, Job Seeker, and Professional—interact with AI-powered modules that provide career guidance. Students can access the AI Career Intelligence Engine to receive tailored career recommendations and use the Skill Insight & Learning Advisor to identify and close skill gaps. Job Seekers use the Smart Resume Enhancer to optimize resumes for ATS compliance, the AI Job Matcher to find suitable job opportunities, and the Skill Insight & Learning Advisor for skill development. Professionals utilize the Growth Pathway Analyzer to explore promotion and upskilling opportunities and the AI Job Matcher to discover advanced career roles. Together, these interactions show how NeuraPath delivers intelligent, data-driven career support to different user groups.

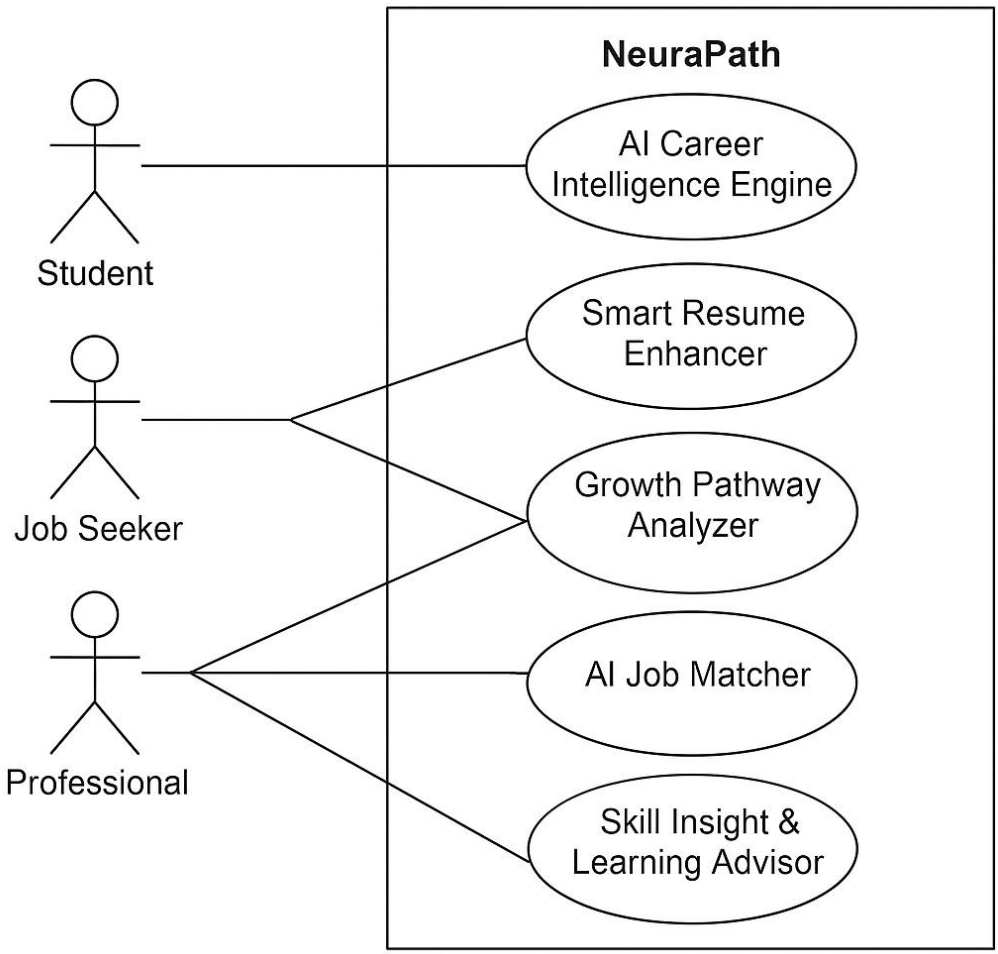


Figure 1: Use Case Diagram

3.2 Use case description

All the use cases are being described below:

Table 1: Use Case 1

Use Case ID:	UC-1
Use Case Name:	AI Career Intelligence Engine
Actors:	**Primary:** Student **Secondary:** AI Career Engine
Description:	A student provides their academic details, skills, and interests. The system uses ML and AI to recommend suitable career paths and identify skill gaps
Trigger:	Student clicks "Get AI Career Intelligence Engine".
Preconditions:	PRE-1: Student must be registered and logged in. PRE-2: Student profile must contain basic academic and skill data.
Postconditions:	POST-**1:** AI Career Intelligence Engine report is generated POST-**2:** Recommended career are stored in the profile for future access. .
Normal Flow:	1. Student logs in. 2. Student selects "AI Career Intelligence Engine". 3. System retrieves student profile. 4. AI analyzes academic data, skills, and market trends. 5. System generates career suggestions. 6. Student views recommendations.
Alternative Flows:	AF-1: If profile is incomplete, the system prompts student to update profile.
Exceptions:	▪ **No Internet Connection** If Student cancels the recommendation process, then NeuraPath terminates the use case. ▪ **Incomplete Profile Data** System detects that required academic or skill information is missing. if Student updates profile information, then NeuraPathrestarts the use case.
Business Rules	BR-**1:** AI Career Intelligence Engines must be aligned with **real-time job market data** retrieved from trusted platforms (Linked-In, Rozee.pk, etc.). BR-**2:** Only **registered Students** can request AI Career Intelligence Engines.
Assumptions:	Student enters accurate academic and skill details.

Table 2: Use Case 2

Use Case ID:	UC-2
Use Case Name:	Smart Resume Enhancer
Actors:	**Primary:** Job Seeker **Secondary:** NLP Parser, ATS Checker
Description:	The job seeker uploads a resume. The system parses it using NLP, checks ATS compliance, and provides optimization suggestions.
Trigger:	Job seeker clicks "Optimize Resume".
Preconditions:	PRE-1: User is logged in as Job Seeker. PRE-2: Resume uploaded in supported format (PDF/DOCX).
Postconditions	POST-1: Smart Resume Enhancer report generated. POST-2: Suggestions stored in user account.
Normal Flow:	1. User selects "Optimize Resume". 2. System prompts for resume upload. 3. User uploads resume. 4. System parses resume with NLP. 5. ATS compliance and keyword analysis performed. 6. Optimization report displayed.
Alternative Flows:	AF-1: If resume file type unsupported, system requests re-upload.
Exceptions:	EX-1: If parsing fails, system displays error.
Business Rules	BR-1: Only PDF and DOCX formats are supported.
Assumptions:	A-1: Resume is written in English.

Table 3: Use Case 3

Use Case ID:	UC-3
Use Case Name:	Growth Pathway Analyzer
Actors:	**Primary:** Professional **Secondary:** Career Analyzer Engine
Description:	A professional provides current job details, salary, and skills. The system analyzes market data to recommend promotion pathways and skill upgrades.
Trigger:	Professional selects "Career Progression".
Preconditions:	PRE-1: Professional is logged in. PRE-2: Profile includes job role and skills.
Postconditions	POST-1: Promotion pathway generated. POST-2: Skill gap report provided.
Normal Flow:	1. Professional logs in. 2. Selects "Growth Pathway Analyzer". 3. System retrieves profile. 4. System compares skills with industry benchmarks. 5. Report generated showing promotion roadmap and learning resources.
Alternative Flows:	If skills not provided, system prompts to add them
Exceptions:	If market data unavailable, system shows "No Data".
Business Rules	Salary benchmarks based on real-time job portals.
Assumptions:	Professional provides valid career details.

Table 4: Use Case 4

Use Case ID:	UC-4
Use Case Name:	AI Job Matcher
Actors:	Primary: Job Seeker (Optional: Professional)
Description:	The system matches user profiles with relevant job opportunities from integrated job platforms using AI-based similarity scoring
Trigger:	User clicks "Find Jobs."
Preconditions:	PRE-1: User is logged in as Job Seeker or Professional. PRE-2: User profile must include key skills and desired role.
Postconditions	POST-1: System displays job results ranked by match score. POST-2: User can view or bookmark job listings.
Normal Flow:	1. User logs in. 2. Selects "Find Jobs." 3. System retrieves user profile. 4. System fetches job data from APIs. 5. Matches jobs based on skills and preferences. 6. Displays top-matched job listings.
Alternative Flows:	AF-1: If no matches found, system suggests similar roles.
Exceptions:	**EX-1: Job API Failure** • If external job platforms fail (e.g., LinkedIn / Rozee.pk), the system shall display previously cached results not older than 48 hours. • If cached data exceeds 48 hours, the system must display: "Live job data is temporarily unavailable. Please try again later." **EX-2: Web Scraper Timeout** • If the scraper exceeds 8 seconds without response, the system must retry once. • If the second attempt fails, switch to cache mode. **EX-3: Partial Data Retrieval** • If only some platforms respond, the system shall: • Display jobs from successful APIs • Show a non-blocking warning: "Some job sources could not be fetched. Results may be incomplete."
Business Rules	BR-1: Job data must be refreshed daily. BR-2: Results must display company, title, and application link.
Assumptions:	User has up-to-date profile information.

Table 5: Use Case 5

Use Case ID:	UC-5
Use Case Name:	Skill Insight & Learning Advisor
Actors:	Primary: Job Seeker(Optional: Student)
Description:	The system analyzes a user's skills and compares them with the requirements of their desired role or career path to identify missing skills.
Trigger:	User selects "Skill Gap Analysis."
Preconditions:	PRE-1: User is logged in. PRE-2: User profile must contain skill data and career goal.
Postconditions	POST-1: System generates a skill gap report. POST-2: Recommended learning resources linked to identified gaps.
Normal Flow:	1. User logs in. 2. Selects "Skill Gap Analysis." 3. System retrieves skills and target role. 4. Compares user profile with job role data. 5. Identifies missing skills and generates report. 6. Displays recommended learning paths.
Alternative Flows:	AF-1: If user has not selected a target role, system prompts to choose one.
Exceptions:	EX-1: If data source unavailable, system shows "Unable to process request."
Business Rules	BR-1: At least 3 missing skills must be displayed for meaningful analysis. BR-2: Recommendations must align with user's current level.
Assumptions:	User provides accurate information about their existing skills

4 Functional Requirements

Table 6: User Onboarding & Smart Profile Management

Field	**FR-1**
Title	User Onboarding & Smart Profile Management
Requirement	The user (Student, Job Seeker, Professional, Admin) shall be able to register and log in using unique credentials. The system shall store and manage user profiles, allowing updates to education, skills, and experience.
Source	Scope Document / End User Need
Rationale	Registration and profiles are required for personalization and role-based access.
Business Rule	BR-1: Each user must have a unique email/phone.
Dependencies	FR-2 (AI Career Intelligence Engines), FR-5 (Resume Upload).
Priority	High

Table 7: AI Career Intelligence Engine

Field	**FR-2**
Title	AI Career Intelligence Engine
Requirement	. The **system shall generate personalized career paths** based on academic details, skills, and interests, aligned with real-time market data.
Source	Scope Document / Student Needs
Rationale	Provides intelligent career guidance tailored to students' backgrounds.
Business Rule	BR-2: At least two career paths must be recommended.
Dependencies	FR-8 (Skill Insight & Learning Advisor).
Priority	High

Table 8: Smart Resume Enhancer

Field	**FR-3**
Title	Smart Resume Enhancer
Requirement	The **Job Seeker shall upload resumes** (PDF/DOCX). The **system shall parse resumes using NLP**, check ATS compliance, and generate improvement suggestions. The system shall only accept resumes that meet the following format requirements: • Supported formats: PDF (text-based) and DOCX • Maximum file size: 5 MB • Minimum length: 1 page, maximum 4 pages • Orientation: Portrait only • Language: English only • Resume must contain selectable text (no scanned images)

	• No password-protected or encrypted files • Images must not exceed 20% of the document area • If a resume is image-only, the system shall display: "*Your resume appears to be scanned. Please upload a text-based PDF or DOCX file.*"
Source	Scope Document / Job Seeker Needs
Rationale	Improves job seekers' chances in ATS-based hiring
Business Rule	BR-5: Only PDF and DOCX formats are accepted.
Dependencies	FR-6 (Job Matching).
Priority	High

Table 9: AI Job Matcher

Field	**FR-4**
Title	AI Job Matcher
Requirement	The **system shall match user profiles** with job postings from external sources (LinkedIn, Rozee.pk) using similarity scoring.
Source	Scope Document / Job Seeker Needs
Rationale	Connects users with real-time job opportunities.
Business Rule	BR-6: Job data must refresh daily.
Dependencies	FR-1 (Profiles), FR-5 (Smart Resume Enhancer)
Priority	High

Table 10: Growth Pathway Analyzer

Field	**FR-5**
Title	Growth Pathway Analyzer
Requirement	The **system shall analyze professional users' career paths** and generate promotion pathways with skill recommendations
Source	Scope Document / Professional Needs
Rationale	Helps professionals identify growth opportunities
Business Rule	BR-7: Salary benchmarks must use updated industry data.
Dependencies	FR-8 (Skill Insight & Learning Advisor).
Priority	Medium

Table 11: Skill Insight & Learning Advisor

Field	**FR-6**
Title	Skill Insight & Learning Advisor
Requirement	The **system shall analyze user skills against job/role requirements** and generate a skill gap report with recommended training
Source	Scope Document / User Needs
Rationale	Helps users identify missing competencies.
Business Rule	BR-8: Report must include at least 3 skills per gap.
Dependencies	FR-2 (AI Career Intelligence Engines)
Priority	High

5 Non-Functional Requirements

This section defines the quality attributes that ensure the NeuraPath system performs efficiently, securely, and reliably across all environments. Each requirement is stated in measurable and verifiable terms.

5.1 Usability

- **USE-1:** At least 85% of first-time users shall be able to complete registration and reach the dashboard within 2 minutes without assistance.
- **USE-2:** The system shall achieve a System Usability Scale (SUS) score ≥ 75, indicating "Good" usability.
- **USE-3:** All user interface elements (buttons, forms, menus) shall maintain consistent styling, verified by a UI audit with 0 critical inconsistencies per screen.
- **USE-4:** Error messages must appear within 1 second of invalid input and must include:
    - Cause of error (e.g., invalid email format)
    - Guidance to fix it

  This will be validated during functional testing.

5.2 Performance

- **PER-1:** The system shall generate the AI Career Intelligence Engine output within 25–35 seconds on a stable 10 Mbps internet connection.
- **PER-2:** The system shall process resume uploads and display optimization results within **20 seconds**.
- **PER-3:** For queries involving extensive market data, the system must display progressive loading stages (profile analysis → market retrieval → model inference → results compilation), with each stage updating the user within 5 seconds.
- **PER-4:** The system shall maintain **99.5% uptime availability** for all core services.
- **PER-5:** The system shall support up to **500 concurrent users** without performance degradation.

5.3 Security

- **SEC-1:** All user data shall be stored securely using **Firebase Authentication** and **Firestore security rules**.
- **SEC-2:** Passwords shall be encrypted using a **one-way hashing algorithm** (e.g., SHA-256).
- **SEC-3:** The system shall use **role-based access control (RBAC)** to restrict unauthorized actions.
- **SEC-4:** All communication between client and server shall use **HTTPS** for data encryption.
- **SEC-5:** Admin functionalities shall be accessible only after **two-factor authentication**.

5.4 Reliability

- **REL-1:** The system shall automatically recover from network disconnections without losing unsaved data.
- **REL-2:** In case of system failure, user data shall remain intact and restorable.
- **REL-3:** The system shall log all critical actions and errors for maintenance tracking.
- **REL-4:** The system shall ensure at least **95% data accuracy** during transactions.

5.5 Maintainability

- **MAI-1:** The codebase shall be modular and follow **MVVM architecture** for easier updates.
- **MAI-2:** Developers shall be able to add or modify modules without affecting existing features.
- **MAI-3:** All source code shall be documented with comments for future maintenance.
- **MAI-4:** Bug fixes and updates shall be deployable without downtime using **Firebase continuous deployment**.

5.6 Portability

- **POR-1:** The system shall run on Android (version 10 and above) and be adaptable for web deployment.
- **POR-2:** The source code shall be platform-independent using **Flutter**.
- **POR-3:** The system shall be easily scalable for iOS version in the future.

6 References

[1] Cloudinary Ltd. (2025). Retrieved from Cloudinary API Documentation: https://cloudinary.com/documentation

[2] Developers, G. (2025). Retrieved from Flutter Official Documentation: https://flutter.dev/docs

[3] Python Software Foundation. (2025). Retrieved from BeautifulSoup and Selenium Official Documentation: https://www.crummy.com/software/BeautifulSoup/ and https://selenium.dev/documentation/